MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 6378

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: TR, Tunable, frequency range 8.49 to 9.6 GHz.

ABSOLUTE RATINGS:

| Parameter: | po (transmitter) | Ebb (open circuit) | Du | Ii | Alt | TA |
|---|---|---|---|---|---|---|
| Unit: | kW | Vdc | ----- | µAdc | ft | °C |
| Maximum: | 30 | ---- | 0.001 | 200 | 10,000 | +100 |
| Minimum: | -- | -500 | ----- | 100 | --- | -40 |

PHYSICAL CHARACTERISTICS:

| | |
|---|---|
| Dimensions: | See outline drawing |
| Weight: | 5 oz. (approx) |

TEST CONDITIONS:

| Parameter: | Ebb | Ii | po | tp | Du | Ri |
|---|---|---|---|---|---|---|
| Unit: | Vdc | µAdc | kW | µs | ------ | MΩ |
| Tolerance: | --- | --- | ±10% | ±10% | ------ | ---- |
| Test 1: | -500 | ---- | ---- | ---- | ------ | 1.25 |
| Test 2: | ---- | 100 | ---- | ---- | ------ | 1.25 |
| Test 3: | -500 | ---- | 30 | 1.0 | 0.001 | 1.25 |
| Test 4: | ---- | 100 | 30 | 0.5 | 0.0005 | 1.25 |
| Test 5: | ---- | 100 | 10 | 0.5 | 0.0005 | 1.25 |

| TEST FREQUENCIES |
|---|
| $F_1$ = 8.490 GHz ±0.1% |
| $F_2$ = 9.375 GHz ±0.1% |
| $F_3$ = 9.600 GHz ±0.1% |

GENERAL:

Qualification - Required

6378

FSC 5960



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| METHOD | REQUIREMENT OR TEST | TEST | CONDITIONS | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | See Note 1 | | | | |
| 4401 | Ignitor ignition time | 1 | | t | --- | 1.0 | sec |
| 4406 | Ignitor voltage drop | 2 | | Eid | 200 | 400 | Vdc |
| 4416 | Insertion loss (1) (Integral cavity) | - | F = $F_2$ | Li | --- | 1.5 | dB |
| 4421 | Ignitor interaction (Insertion loss) | 2 | | ΔLi | --- | 0.2 | dB |
| 4426 | Tuning range | - | F = $F_1$ to $F_3$ | --- | 5 | --- | turns |
| 4452 | Leakage power | 5 | F = $F_2$, T = 25°C | $\frac{Po}{Du}$ | --- | 30 | mW |
| 1027 | Temperature cycling (nonoperating) | - | | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | |
| 1031 | High-frequency vibration | - | t = 15 min | --- | --- | --- | --- |
| 4461 | Loaded Q | - | | QL | --- | 350 | --- |
| 4016 | Vibration detuning | - | F = $F_2$, t = 12 hours | ΔF | --- | ±3 | MHz |
| 4411 | Ignitor oscillation (obsolescent) | - | | Li | --- | 60 | μAdc |
| 4416 | Insertion loss (2) (integral cavity) | - | F = $F_1$ and $F_3$ | Li | --- | 1.5 | dB |
| 4466 | Temperature coefficient of frequency | - | F = $F_1$ See Note 2 | ΔF | --- | -20 | MHz |
| 4471 | Recovery time | 4 | F = $F_2$ | t | --- | 4.0 | μs |
| 4002 | Pressurizing | - | 45 $lb_f/in^2$ | --- | --- | --- | --- |



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| METHOD | REQUIREMENT OR TEST | TEST | CONDITIONS | SYMBOL | LIMITS MIN | LIMITS MAX | UNITS |
|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 3 | | | | | | |
| 4551 | Life test | 3 | F = $F_2$, Group D | t | 500 | --- | hrs |
| ---- | Life test end points: | | | | | | |
| 4406 | Ignitor voltage drop | 2 | | Eid | --- | 450 | Vdc |
| 4416 | Insertion loss (1) | - | F = $F_2$ | Li | --- | 2.0 | dB |
| 4421 | Ignitor interaction | 2 | | Li | --- | 0.5 | mW |
| 4452 | Leakage power | 5 | F = $F_2$, T = 25°C | $\frac{Po}{Du}$ | --- | 35 | mW |
| 4471 | Recovery time | 4 | F = $F_2$ | t | --- | 8.0 | µs |
| 1027 | Temperature cycling | - | Group D | --- | 10 | --- | cycles |

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1 shall be 1.0 percent, inspection level II.
2. The frequency drift shall be measured with no adjustment of the tuning mechanism.

Custodians:

Army - EL
Navy - EC
Air Force - 85

Review activities:

Army - MI
Air Force - 99

User activities:

Army - AR
Air Force - 11

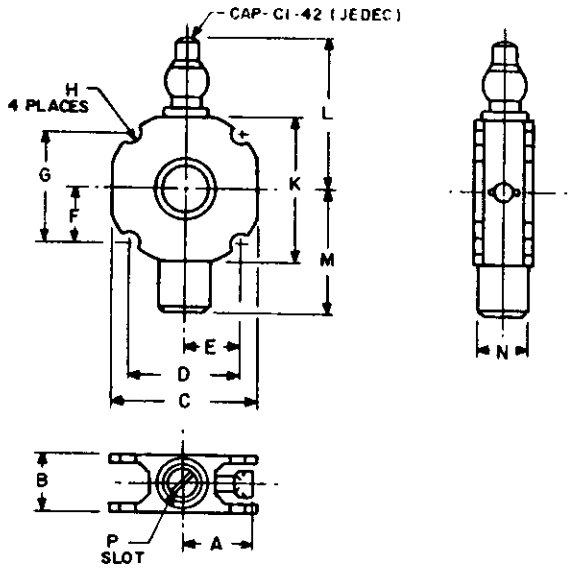
Preparing activity:

Air Force - 85

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| DIMENSIONS | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| LTR | INCHES | | MM | | LTR | INCHES | | MM | |
| | MIN | MAX | MIN | MAX | | MIN | MAX | MIN | MAX |
| QUALITY CONFORMANCE INSPECTION, PART 1 | | | | | M | – | 1.422 | – | 36.12 |
| A | – | .750 | – | 19.05 | N | – | .530 DIA | – | 13.46 DIA |
| B | .607 | .617 | 15.42 | 15.67 | QUALITY CONFORMANCE INSPECTION, PART 3 | | | | |
| L | – | 1.625 | – | 41.28 | C | 1.610 | 1.640 | 40.89 | 41.66 |
| QUALITY CONFORMANCE INSPECTION, PART 2 | | | | | E | .608 | .612 | 15.44 | 15.54 |
| D | 1.216 | 1.224 | 30.89 | 31.09 | F | .638 | .642 | 16.21 | 16.31 |
| G | 1.276 | 1.284 | 32.41 | 32.61 | K | 1.610 | 1.640 | 40.89 | 41.66 |
| H | .085 RAD | .095 RAD | 2.16 R | 2.41 R | P | .031 | .062 | .79 | 1.57 |

NOTES:
0. EXHAUST TUBE SHALL NOT EXTEND BEYOND FLANGE.

FIGURE 1. Outline drawing



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